Proofs for *Reprise: One Graph Pass for Two GNN Updates*

Notation

To define the frozen gradients, fix the context $\mathcal{C}$ and the shared Body parameters $\theta \in \mathbb{R}^p$. Write $z = f_\theta(\mathcal{C})$ for the Body output and $J = \partial z/\partial\theta$ for its Jacobian. The two Head states ϕ_0, ϕ_1 produce the feedbacks $\lambda_i = \partial_z L(z, \phi_i)$ and the frozen Body gradients

$$g_i = J^\top \lambda_i, \qquad q = \tfrac{1}{2}(g_0 + g_1), \qquad d = \tfrac{1}{2}(g_1 - g_0), \tag{A.1}$$

so that $g_0 = q - d$ and $g_1 = q + d$. Both gradients use the same J, so one backward pass of the averaged feedback returns their mean exactly:

$$J^\top \tfrac{1}{2}(\lambda_0 + \lambda_1) = \tfrac{1}{2}(J^\top \lambda_0 + J^\top \lambda_1) = q.$$

The measured Body directions are the columns of $N \in \mathbb{R}^{p\times k}$. Each column is a vector in the full Body parameter space: a direction that acts on a few parameters, such as the scale and offset of one channel, has zeros in every other coordinate. The two Head queries report $\mu_i = N^\top g_i$, and their half-difference is $s = \frac{1}{2}(\mu_1 - \mu_0) = N^\top d$. With $\dagger$ denoting the Moore–Penrose pseudoinverse, define the projector Π, correction r, and unmeasured part e by

$$\Pi = N(N^\top N)^\dagger N^\top, \qquad r = N(N^\top N)^\dagger s = \Pi d, \qquad e = (I - \Pi)d. \tag{A.2}$$

Reprise submits $q - r$ and then $q + r$ to the Body optimizer. Inner products and norms are Euclidean, and matrix contractions use the Frobenius inner product.

Proposition 1 (Measurement by symmetry)**.** *Fix the context and the parameters, let F be the logits, and let n be a Body direction. A single function of the Head gradient $\nabla_\phi L$ returns $n^\top \nabla_\theta L$ for every differentiable loss $L = \ell(F)$ if and only if some Head direction m satisfies $D_\theta F\, n + D_\phi F\, m = 0$. Then $n^\top \nabla_\theta L = -m^\top \nabla_\phi L$.*

Proof. Flatten the logits into a vector in $\mathbb{R}^\ell$, and at the fixed context and parameters write

$$A = D_\theta F, \qquad B = D_\phi F, \qquad a = \nabla_F \ell.$$

The chain rule expresses both gradients through the same logit gradient:

$$\nabla_\theta L = A^\top a, \qquad \nabla_\phi L = B^\top a, \qquad n^\top \nabla_\theta L = (An)^\top a. \tag{A.3}$$

Every $a \in \mathbb{R}^\ell$ arises from some differentiable loss, namely the linear loss $\ell_a(F) = a^\top F$.

Sufficiency. If $An + Bm = 0$, then for every loss

$$n^\top \nabla_\theta L = a^\top An = -a^\top Bm = -m^\top \nabla_\phi L,$$

so the linear function $h \mapsto -m^\top h$ returns the Body directional derivative from the Head gradient.

Necessity. Suppose that no Head direction m satisfies $An + Bm = 0$, that is, $An \notin \operatorname{range} B$. Let P_B be the orthogonal projector onto $\operatorname{range} B$ and set $v = (I - P_B)An \neq 0$. Then $B^\top v = 0$, and since $P_B An$ is orthogonal to v,

$$v^\top An = v^\top (P_B An + v) = \|v\|^2 > 0.$$

The losses $\ell_0(F) = 0$ and $\ell_1(F) = v^\top F$ have the same Head gradient, $B^\top 0 = 0 = B^\top v$, but different Body directional derivatives, 0 and $\|v\|^2$. Any function of the Head gradient, linear or not, assigns both losses the same value and is therefore wrong for at least one of them. Recovery for every differentiable loss thus forces $An \in \operatorname{range} B$, which provides the required m. □

The channel contraction. Let the linear Head have logits $ZW^\top + b$, where $Z \in \mathbb{R}^{n\times h}$ is the Body output and $W \in \mathbb{R}^{c\times h}$, and let $A \in \mathbb{R}^{n\times c}$ now denote the derivative of the loss with respect to the logits. The Head's backward pass produces

$$G = \nabla_W L = A^\top Z, \qquad \lambda = AW,$$

the second being the feedback to the Body output. Suppose the last Body parameters act on channel j through a learned scale and offset, $\gamma_j \hat{x}_{vj} + \beta_j$ at node v, and that the operations T that follow commute with positive channel scaling: $T(HD) = T(H)D$ for every positive diagonal D. Multiplying (γ_j, β_j) by $\alpha > 0$ multiplies column j of the affine output by α and hence multiplies $Z_{:,j}$ by α. Dividing $W_{:,j}$ by α then leaves $ZW^\top$, and with it every logit, unchanged.

The logits are constant along this one-parameter family, so differentiating at $\alpha = 1$ gives $D_\theta F\, n_j + D_\phi F\, m_j = 0$. Here the Body direction n_j has entries (γ_j, β_j) at the scale and offset of channel j and zeros elsewhere, and the Head direction m_j has entries $-W_{:,j}$ in column j of W and zeros elsewhere. Writing g_{γ_j} and g_{β_j} for the entries of $\nabla_\theta L$ at γ_j and β_j, Proposition 1 gives the channel contraction

$$\begin{aligned}\mu_j = \gamma_j g_{\gamma_j} + \beta_j g_{\beta_j} &= n_j^\top \nabla_\theta L = -m_j^\top \nabla_\phi L = W_{:,j}^\top G_{:,j} \\ &= \sum_c W_{cj} \sum_v A_{vc} Z_{vj} = \sum_v Z_{vj} \sum_c A_{vc} W_{cj} = \sum_v Z_{vj} \lambda_{vj}. \end{aligned} \tag{A.4}$$

The Head thus measures the Body gradient along n_j with a single inner product between a weight column and its own gradient, and the last expression identifies the same number as the contraction of Body output channel j with its feedback. For an ordinary affine channel $w_j^\top x + b_j$, scaling the weight row and the bias together gives the same argument with $n_j = (w_j, b_j)$. At Head query i, the contraction pairs the pre-update weights W_i with their gradient G_i. Both belong to the Head state ϕ_i at which g_i is defined, so the query reports $n_j^\top g_i$.

Theorem 2 (Exact recovery)**.** *The pair $(q-r, q+r)$ has mean q, satisfies $N^\top(q-r) = N^\top g_0$ and $N^\top(q+r) = N^\top g_1$, and among all pairs with these properties uniquely minimizes $\|x_0 - q\|^2 + \|x_1 - q\|^2$. Its squared error to the frozen pair is $E_{\mathrm{R}} = E_{\mathrm{M}} - 2\|r\|^2$, where $E_{\mathrm{M}} = 2\|d\|^2$ is the error of the mean pair (q, q); reversing the pair gives $E_{\mathrm{R}} + 8\|r\|^2$. Among all pairs, it uniquely maximizes the guaranteed reduction of this error below E_{M}, the smallest reduction over all frozen pairs with the same q, N, and s, and this guaranteed reduction is $2\|r\|^2$.*

Proof. Measured and unmeasured parts. If $N \neq 0$ has rank ρ, take a compact singular value decomposition $N = U_\rho \Sigma_\rho V_\rho^\top$, where U_ρ and V_ρ have orthonormal columns and Σ_ρ is positive diagonal. Then

$$\begin{aligned}(N^\top N)^\dagger &= V_\rho \Sigma_\rho^{-2} V_\rho^\top, \\ \Pi &= U_\rho \Sigma_\rho V_\rho^\top \, V_\rho \Sigma_\rho^{-2} V_\rho^\top \, V_\rho \Sigma_\rho U_\rho^\top = U_\rho U_\rho^\top.\end{aligned}$$

Hence Π is the orthogonal projector onto range N, with $\ker \Pi = \ker N^\top$ and $N^\top \Pi = N^\top$. If $N = 0$, then $\Pi = 0$ and the same statements hold. In either case

$$d = r + e, \qquad r \in \operatorname{range} N, \qquad e \in \ker N^\top, \qquad \langle r, e \rangle = 0, \tag{A.5}$$

and $N^\top r = N^\top d = s$.

Mean and measurements. The pair $(q-r, q+r)$ has mean q, and

$$N^\top(q-r) = N^\top(q-d) = N^\top g_0, \qquad N^\top(q+r) = N^\top(q+d) = N^\top g_1.$$

Least change. Every pair with mean q has the form $(x_0, x_1) = (q-u, q+u)$. It matches both measurements exactly when $N^\top u = s = N^\top r$, that is, when $u = r + w$ with $w \in \ker N^\top$. Since $r \in \operatorname{range} N$ is orthogonal to w,

$$\|x_0 - q\|^2 + \|x_1 - q\|^2 = 2\|u\|^2 = 2\|r\|^2 + 2\|w\|^2,$$

which is smallest exactly when $w = 0$, that is, for $(q-r, q+r)$. When the measured directions have disjoint supports, $N^\top N$ is diagonal and the decoder separates into one scalar problem per direction, $r = \sum_j (s_j/\|n_j\|^2)\, n_j$ over the nonzero directions. For a channel direction this reads

$$r_j = \frac{s_j}{\gamma_j^2 + \beta_j^2}\,(\gamma_j, \beta_j).$$

A zero direction has $s_j = n_j^\top d = 0$, and the pseudoinverse assigns it no correction.

Errors of recovery, the mean pair, and reversal. Every ordered pair has a unique representation $(c-u, c+u)$ by its mean c and half-difference u. Its squared error to the frozen pair $(q-d, q+d)$ is

$$\begin{aligned} E(c, u; d) &= \|(c-q) - (u-d)\|^2 + \|(c-q) + (u-d)\|^2 \\ &= 2\|c-q\|^2 + 2\|u-d\|^2, \end{aligned} \tag{A.6}$$

since the two cross terms cancel. With (A.5),

$$\begin{aligned} E_{\mathrm{M}} &= E(q, 0; d) = 2\|d\|^2 = 2\|r\|^2 + 2\|e\|^2, \\ E_{\mathrm{R}} &= E(q, r; d) = 2\|e\|^2 = E_{\mathrm{M}} - 2\|r\|^2, \\ E_{\mathrm{rev}} &= E(q, -r; d) = 2\|2r + e\|^2 = 8\|r\|^2 + 2\|e\|^2 = E_{\mathrm{R}} + 8\|r\|^2. \end{aligned}$$

Recovery removes the measured part of the mean pair's error, and reversal adds four times that amount to the error of recovery.

The guaranteed improvement. The observations q, N, and s determine the frozen half-difference up to its unmeasured part. The compatible half-differences are exactly

$$\{d : N^\top d = s\} = \{r + e : e \in \ker N^\top\},$$

because every $r + e$ in the right-hand set has measurement s, and any compatible d differs from r by a vector in $\ker N^\top$. For an arbitrary pair $(c-u, c+u)$, define its guaranteed improvement over the mean pair as the worst case over all compatible frozen pairs,

$$\Gamma(c, u) = \inf_{e \in \ker N^\top} \Big[E_{\mathrm{M}}(r+e) - E(c, u; r+e) \Big],$$

where $E_{\mathrm{M}}(d) = E(q, 0; d) = 2\|d\|^2$. Expanding (A.6) gives

$$E_{\mathrm{M}}(r+e) - E(c, u; r+e) = 4\langle r+e, u\rangle - 2\|u\|^2 - 2\|c-q\|^2. \tag{A.7}$$

Split $u = \Pi u + w$ with $w = (I - \Pi)u \in \ker N^\top$. If $w \neq 0$, the compatible choice $e = -tw$ with $t > 0$ gives $\langle e, u\rangle = -t\|w\|^2$, so (A.7) tends to $-\infty$ as $t \to \infty$ and $\Gamma(c, u) = -\infty$. If $w = 0$, then $u \in \operatorname{range} N$ and $\langle e, u\rangle = 0$ for every compatible e, so (A.7) no longer depends on e. Completing the square, $4\langle r, u\rangle - 2\|u\|^2 = 2\|r\|^2 - 2\|u - r\|^2$, and therefore

$$\Gamma(c, u) = 2\|r\|^2 - 2\|u - r\|^2 - 2\|c - q\|^2.$$

Its largest value, $2\|r\|^2$, is attained only at $c = q$ and $u = r$. □

The maximization ranges over all ordered pairs, including pairs that move the mean or disregard the measurements. Keeping q, matching the measurements, and correcting nowhere else therefore emerge as exactly what the largest guarantee requires.

Proposition 3 (Order in Adam's momentum)**.** *Without weight decay or clipping and from a common incoming state, let $\delta m_X = m_2(d) - m_2(x_X)$ be the block-end first moment of the frozen pair minus that of an execution that submits $q - x_X$, then $q + x_X$. For Reprise ($x_{\mathrm{R}} = r$), the mean pair ($x_{\mathrm{M}} = 0$), and the reversed correction ($x_{\mathrm{rev}} = -r$),*

$$\delta m_{\mathrm{R}} = (1-b_1)^2(I-\Pi)d, \qquad \delta m_{\mathrm{M}} = (1-b_1)^2 d, \qquad \delta m_{\mathrm{rev}} = (1-b_1)^2(d+r).$$

Hence

$$\|\delta m_{\mathrm{M}}\|^2 - \|\delta m_{\mathrm{R}}\|^2 = (1-b_1)^4\|r\|^2, \qquad \|\delta m_{\mathrm{rev}}\|^2 - \|\delta m_{\mathrm{R}}\|^2 = 4(1-b_1)^4\|r\|^2.$$

Proof. Let m be the common incoming first moment. For the ordered inputs $q - x$ and then $q + x$, Adam's first-moment recurrence gives

$$\begin{aligned} m_1(x) &= b_1 m + (1-b_1)(q-x), \\ m_2(x) &= b_1 m_1(x) + (1-b_1)(q+x) \\ &= b_1^2 m + (1-b_1)(1+b_1)\, q + (1-b_1)(1-b_1)\, x \\ &= b_1^2 m + (1-b_1^2)\, q + (1-b_1)^2\, x. \end{aligned}$$

The frozen pair is the case $x = d$. Subtracting a candidate from it cancels the incoming state and the mean:

$$\delta m_1(x) = m_1(d) - m_1(x) = (1-b_1)(x-d), \qquad \delta m_2(x) = m_2(d) - m_2(x) = (1-b_1)^2(d-x). \quad \text{(A.8)}$$

With $d = r + e$, the block-end errors of the three executions are

$$\delta m_{\mathrm{R}} = (1-b_1)^2 e, \qquad \delta m_{\mathrm{M}} = (1-b_1)^2(r+e), \qquad \delta m_{\mathrm{rev}} = (1-b_1)^2(2r+e),$$

which are the stated expressions, since $e = (I-\Pi)d$ and $2r + e = d + r$. Because r and e are orthogonal,

$$\begin{aligned} \|\delta m_{\mathrm{M}}\|^2 - \|\delta m_{\mathrm{R}}\|^2 &= (1-b_1)^4(\|r+e\|^2 - \|e\|^2) = (1-b_1)^4\|r\|^2, \\ \|\delta m_{\mathrm{rev}}\|^2 - \|\delta m_{\mathrm{R}}\|^2 &= (1-b_1)^4(\|2r+e\|^2 - \|e\|^2) = 4(1-b_1)^4\|r\|^2. \end{aligned}$$

For Reprise, (A.8) gives $\delta m_1(r) = -(1-b_1)e$ and $\delta m_2(r) = (1-b_1)^2 e$. Both lie in $\ker N^\top$, so $N^\top \delta m_1(r) = N^\top \delta m_2(r) = 0$. After each of the two updates, Reprise's first moment therefore agrees with the frozen pair's along every measured direction. With a common scalar clock, bias correction multiplies the first moments of both executions by the same factor $1/(1-b_1^t)$, so the corrected moments agree along the measured directions as well. □

The second moment expands in the same way. From a common incoming v,

$$\begin{aligned} v_2(x) &= b_2^2 v + (1-b_2)\big[b_2(q-x)^{\odot 2} + (q+x)^{\odot 2}\big] \\ &= b_2^2 v + (1-b_2^2)(q^{\odot 2} + x^{\odot 2}) + 2(1-b_2)^2\, q \odot x, \end{aligned}$$

with elementwise squares and products. The term $x^{\odot 2}$ carries the energy of the change, and the term $q \odot x$ carries its order. Recovery and reversal therefore share the same squared-change energy $r^{\odot 2}$ and carry ordered terms of opposite sign, whereas the mean pair carries neither.

Displacement. Fix the mean q, the initial Body parameters, the incoming optimizer state, the clock, and all schedules. For a half-difference x, let $U(x)$ be the net displacement of the Body parameters after the optimizer consumes $q - x$ and then $q + x$. The mean pair, Reprise, reversal, and the frozen pair produce $U(0)$, $U(r)$, $U(-r)$, and $U(d)$.

Proposition 4 (From gradients to motion)**.** *Let $e = d - r$.* (i) *If $U(x) = U(0) + Kx$ is affine, Reprise reduces the squared displacement error of the mean pair by $a + 2t$ and reversal increases Reprise's by $4a + 4t$, where $a = \|Kr\|^2$ and $t = \langle Kr, Ke\rangle$; the reduction is nonnegative for every d if and only if $\Pi K^\top K(I - \Pi) = 0$.*
(ii) *If the optimizer acts coordinatewise and Q is a coordinate projector with $Qr = r$, then from the common incoming state the pair $q \mp Qd$ reproduces the frozen pair's parameters and optimizer state on the coordinates of Q after each of the two updates and agrees with Reprise outside them, so $\|U(d) - U(Qd)\| \le \|U(d) - U(r)\|$.*

Proof. Affine displacement errors. Assume $U(x) = U(0) + Kx$. Measured from the frozen displacement $U(d)$, the errors of the mean pair, Reprise, and reversal are

$$U(d) - U(0) = K(r + e), \qquad U(d) - U(r) = Ke, \qquad U(d) - U(-r) = K(2r + e).$$

With $a = \|Kr\|^2$ and $t = \langle Kr, Ke\rangle$, expanding the squared norms gives

$$\|U(d) - U(0)\|^2 - \|U(d) - U(r)\|^2 = \|Kr + Ke\|^2 - \|Ke\|^2 = a + 2t,$$
$$\|U(d) - U(-r)\|^2 - \|U(d) - U(r)\|^2 = \|2Kr + Ke\|^2 - \|Ke\|^2 = 4a + 4t.$$

Sufficiency of the cross-block condition. Write $M = K^\top K$. Since $r = \Pi d$ and $e = (I - \Pi)d$,

$$t = r^\top M e = d^\top \Pi M (I - \Pi)\, d.$$

If $\Pi M(I - \Pi) = 0$, then $t = 0$ for every d, the reduction equals $\|Kr\|^2 \ge 0$, and reversal's increase is exactly four times this reduction, as for the first moment in Proposition 3.

Necessity. Suppose $\Pi M(I - \Pi) \ne 0$, and pick a vector ω with $\Pi M(I - \Pi)\omega \ne 0$. Then $e_0 = (I - \Pi)\omega$ lies in $\ker \Pi = \ker N^\top$ and $h = \Pi M e_0 \ne 0$. Put $r_0 = h$, which lies in $\operatorname{range} \Pi = \operatorname{range} N$. Because Π is symmetric and $\Pi h = h$,

$$r_0^\top M e_0 = h^\top \Pi M e_0 = \|h\|^2 > 0.$$

For $d_\tau = r_0 - \tau e_0$ with $\tau > 0$, the measured part is $\Pi d_\tau = r_0$ and the unmeasured part is $-\tau e_0$, so the reduction becomes

$$a + 2t = \|Kr_0\|^2 - 2\tau\|h\|^2,$$

which is negative once $\tau > \|Kr_0\|^2/(2\|h\|^2)$. A nonnegative reduction for every d therefore requires $\Pi K^\top K(I - \Pi) = 0$.

Coordinatewise inputs. Let Q be a diagonal coordinate projector with $Qr = r$. The pair $q \mp Qd$ submits $q_j - d_j$ and then $q_j + d_j$ on every coordinate j selected by Q, which are exactly the frozen inputs. On a coordinate outside Q, $(Qd)_j = 0$ and $r_j = (Qr)_j = 0$, so the pair submits q_j twice, exactly as Reprise does there.

Equality of parameters and optimizer state. A coordinatewise optimizer updates each coordinate by a deterministic recurrence

$$(\theta_j^{(t+1)}, \xi_j^{(t+1)}) = \mathcal{A}_t(\theta_j^{(t)}, \xi_j^{(t)}, g_j^{(t)}),$$

where ξ_j collects the coordinate's optimizer state and $\mathcal{A}_t$ may depend on the common clock and scalar schedules.

All executions start from the same parameters and state. On the coordinates of Q, the first inputs agree with the frozen pair's, so parameters and state agree after the first update. The second inputs agree as well, so they also agree after the second update. Outside Q, the same induction compares the pair with Reprise. Consequently

$$Q(U(d) - U(Qd)) = 0, \qquad (I - Q)(U(d) - U(Qd)) = (I - Q)(U(d) - U(r)).$$

Because Q is an orthogonal projector,

$$\|U(d) - U(Qd)\|^2 = \|(I - Q)(U(d) - U(r))\|^2 \le \|U(d) - U(r)\|^2,$$

and taking square roots proves (ii). □

Full-layer recovery. Let Q be a coordinate projector that selects a layer containing the measured directions, so that $QN = N$. Full-layer recovery adds to r the layer's share of the unmeasured change, $e_Q = (Q - \Pi)d$, which satisfies $N^\top e_Q = N^\top d - N^\top d = 0$. The pairs $q \mp (r + \alpha e_Q)$, $0 \le \alpha \le 1$, keep the mean and every measurement and move from Reprise at $\alpha = 0$ to full-layer recovery at $\alpha = 1$. In Proposition 5, $Q = I$, so $e_Q = d - r$ and $\alpha = 1$ gives the frozen pair.

Proposition 5 (Fidelity does not rank updates)**.** *Take one input whose normalized activation is* 1*, a Body with one learned scale and offset, $z = \gamma + \beta$, a scalar Head w, and loss $\frac{1}{2}(wz - y)^2$ with $y = \frac{1}{2}$, starting at $(\gamma, \beta, w) = (1, 0, 1)$. Starting both optimizers from zero moments and clocks, without weight decay or clipping, the Head takes two native Adam steps ($b_1 = 0.9$, $b_2 = 0.999$, $\epsilon = 10^{-8}$) at the shared $z = 1$; the Body then receives $q \mp (r + \alpha e_Q)$ for the radial direction $(1, 0)$ and $Q = I$. For every common Head and Body learning rate $\eta \in (0, \frac{1}{10}]$, every execution lowers the initial loss, and as α rises from 0 to 1:*
(i) *the error to the frozen pair and the displacement error $\|U(d) - U(r + \alpha e_Q)\|$ of Proposition 4 decrease strictly to zero;*
(ii) *the loss after the block increases strictly if and only if Reprise's Body output stops at or short of y/w_2, the output that minimizes this loss for the updated Head weight w_2, which holds for every $\eta \le (3 - \sqrt{5})/8$, whereas at $\eta = \frac{1}{10}$ the loss decreases strictly.*

Proof. Head steps and frozen Body gradients. At the shared output $z = 1$ the loss is $\frac{1}{2}(w - y)^2$ as a function of w, so the first Head gradient is $h_0 = w_0 - y = \frac{1}{2}$. From zero moments and clock, Adam's first bias-corrected moments are $\hat{m}_1 = h_0$ and $\hat{v}_1 = h_0^2$, so

$$w_1 = 1 - \eta \, \frac{1/2}{1/2 + \epsilon}. \tag{A.9}$$

For $0 < \eta \le \frac{1}{10}$ this gives $0.9 < w_1 < 1$, and the second Head gradient $h_1 = w_1 - \frac{1}{2}$ satisfies $0.4 < h_1 < 0.5$. Since $z = \gamma + \beta$, the Body gradient at Head state i has two equal entries,

$$g_i = w_i(w_i - y)\,(1, 1) = a_i(1, 1),$$

with $a_0 = \frac{1}{2}$ and $0.36 < a_1 < \frac{1}{2}$, because $w \mapsto w(w - \frac{1}{2})$ is increasing on $[0.9, 1]$. Set

$$\bar{q} = \tfrac{1}{2}(a_0 + a_1) > 0.43, \qquad u = \tfrac{1}{2}(a_0 - a_1) > 0, \qquad s_0 = u/\bar{q} \in (0, 1). \tag{A.10}$$

Then $q = \bar{q}(1,1)$ and $d = -u(1,1)$. The radial direction at $(\gamma, \beta) = (1,0)$ is $N = (1,0)^\top$, so

$$\Pi = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad r = (-u, 0), \qquad e_Q = d - r = (0, -u).$$

The pair $q \mp (r + \alpha e_Q)$ therefore submits

	first input	second input
γ	$\bar{q} + u$	$\bar{q} - u$
β	$\bar{q} + \alpha u$	$\bar{q} - \alpha u$

$(0 \le \alpha \le 1)$, (A.11)

and all of these inputs are positive because $u < \bar{q}$.

Two cold Adam steps. Consider a scalar coordinate that starts from zero moments and clock and receives $g_0 = \bar{g}(1+s)$ and then $g_1 = \bar{g}(1-s)$, where $\bar{g} > 0$ and $0 \le s < 1$. The first bias-corrected moments are $\hat{m}_1 = g_0$ and $\hat{v}_1 = g_0^2$, so the first step moves the coordinate by

$$-\eta \frac{g_0}{g_0 + \epsilon} = -\eta \frac{1+s}{1+s+k}, \qquad k = \epsilon/\bar{g}.$$

At the second step,

$$b_1 g_0 + g_1 = \bar{g}\big[(1+b_1) - (1-b_1)s\big],$$
$$b_2 g_0^2 + g_1^2 = \bar{g}^2\big[(1+b_2) - 2(1-b_2)s + (1+b_2)s^2\big],$$

so the bias-corrected moments are

$$\hat{m}_2 = \frac{(1-b_1)(b_1 g_0 + g_1)}{1 - b_1^2} = \frac{b_1 g_0 + g_1}{1 + b_1} = \bar{g}(1 - As),$$
$$\hat{v}_2 = \frac{(1-b_2)(b_2 g_0^2 + g_1^2)}{1 - b_2^2} = \frac{b_2 g_0^2 + g_1^2}{1 + b_2} = \bar{g}^2(1 - 2Bs + s^2),$$

where

$$A = \frac{1-b_1}{1+b_1} = \frac{1}{19}, \qquad B = \frac{1-b_2}{1+b_2} = \frac{1}{1999}.$$

Adam adds ϵ outside the square root, so the second step moves the coordinate by $-\eta\hat{m}_2/(\sqrt{\hat{v}_2} + \epsilon)$, and the two steps together move it by $-\eta T_k(s)$, where

$$T_k(s) = \frac{1+s}{1+s+k} + \frac{1 - As}{\sqrt{1 - 2Bs + s^2} + k}. \tag{A.12}$$

For the Body, $\bar{g} = \bar{q} > 0.43$ and $k = \epsilon/\bar{q} < 10^{-7}$.

T_k is strictly decreasing. Put $D(s) = \sqrt{1 - 2Bs + s^2}$, which is positive since $B < 1$, and note that $D'(s) = (s - B)/D(s)$. Since $D^2 = 1 - 2Bs + s^2$,

$$AD^2 + (1 - As)(s - B) = A - 2ABs + As^2 + s - B - As^2 + ABs = (A - B) + (1 - AB)s,$$

and therefore

$$\frac{d}{ds}\frac{1 - As}{D + k} = -\frac{AD(D+k) + (1 - As)(s - B)}{D(D+k)^2} = -\frac{(A - B) + (1 - AB)s + AkD}{D(D+k)^2}.$$

Hence

$$T_k'(s) = \frac{k}{(1+s+k)^2} - \frac{(A - B) + (1 - AB)s + AkD}{D(D+k)^2}. \tag{A.13}$$

For $s \geq 0$ we have $D^2 \leq (1+s)^2$, so $D \leq 1+s$. Moreover $(1-AB)-(A-B) = (1-A)(1+B) > 0$, so

$$(A-B) + (1-AB)s \geq (A-B)(1+s) \geq (A-B)D.$$

The numerator of the second fraction in (A.13) is therefore at least $(A-B+Ak)D$, and since $A-B+Ak > 0$ and $D+k \leq 1+s+k$,

$$T_k'(s) \leq \frac{k}{(1+s+k)^2} - \frac{A-B+Ak}{(D+k)^2} \leq \frac{(1-A)k-(A-B)}{(1+s+k)^2} < 0 \tag{A.14}$$

whenever

$$k < \frac{A-B}{1-A} = \frac{110}{1999},$$

which holds with a wide margin for $k < 10^{-7}$. For $0 \leq s < 1$ both terms of $T_k(s)$ are positive, and $T_k(0) = 2/(1+k) < 2$, so

$$0 < T_k(s) < 2 \qquad (0 \leq s < 1). \tag{A.15}$$

Body outputs and the two fidelity errors. Since $\bar{q} \pm u = \bar{q}(1 \pm s_0)$ and $\bar{q} \pm \alpha u = \bar{q}(1 \pm \alpha s_0)$, applying (A.12) to the inputs (A.11) gives

$$\gamma_\alpha = 1 - \eta T_k(s_0), \qquad \beta_\alpha = -\eta T_k(\alpha s_0), \qquad z_\alpha = 1 - \eta\big[T_k(s_0) + T_k(\alpha s_0)\big]. \tag{A.16}$$

By (A.14), $\partial z_\alpha / \partial \alpha = -\eta s_0 T_k'(\alpha s_0) > 0$ on $[0,1]$: a more faithful pair moves the Body output less.

At $\alpha = 1$ the pair is the frozen pair, because $r + e_Q = d$. For general α its squared gradient error is

$$E_\alpha = 2\|d - (r + \alpha e_Q)\|^2 = 2(1-\alpha)^2\|e_Q\|^2 = 2u^2(1-\alpha)^2.$$

The γ coordinate receives the frozen inputs for every α, so only β contributes to the displacement error,

$$\|U(d) - U(r + \alpha e_Q)\| = \eta\big[T_k(\alpha s_0) - T_k(s_0)\big].$$

Both errors decrease strictly to zero as α rises from 0 to 1, which proves (i).

Every execution lowers the loss. The Head's inputs $h_0 = \frac{1}{2}$ and $h_1 \in (0.4, 0.5)$ are positive and decreasing, so they have the form used above with

$$\bar{h} = \tfrac{1}{2}(h_0 + h_1) > 0.45, \qquad s_h = \frac{h_0 - h_1}{h_0 + h_1} \in (0,1), \qquad k_h = \epsilon/\bar{h} < 10^{-7}.$$

Hence $w_2 = 1 - \eta T_{k_h}(s_h)$, and (A.15) gives $1 - 2\eta < w_2 < 1$. Applied to (A.16), the same bound gives $1 - 4\eta < z_\alpha < 1$. For $0 < \eta \leq 0.1$ these imply $w_2 > 0.8$ and $z_\alpha > 0.6$, so $0.48 < w_2 z_\alpha < 1$ and $|w_2 z_\alpha - \frac{1}{2}| < \frac{1}{2}$. Every execution therefore ends with

$$L_\alpha = \tfrac{1}{2}\big(w_2 z_\alpha - \tfrac{1}{2}\big)^2 < \tfrac{1}{8},$$

below the initial loss $\frac{1}{2}(1 \cdot 1 - \frac{1}{2})^2 = \frac{1}{8}$.

When the loss increases with α. The final Head weight $w_2 > 0$ is common to all executions, and

$$\frac{\partial L_\alpha}{\partial \alpha} = w_2\,(w_2 z_\alpha - y)\,\frac{\partial z_\alpha}{\partial \alpha}. \tag{A.17}$$

The first and last factors are positive, so the derivative has the sign of $z_\alpha - y/w_2$, where y/w_2 is the Body output that minimizes the loss for the updated Head. Because z_α increases strictly, its smallest value is Reprise's output z_0. If $z_0 \geq y/w_2$, then $z_\alpha > y/w_2$ for every $\alpha > 0$, so the

derivative is positive on $(0, 1]$. It can vanish only at $\alpha = 0$, when $z_0 = y/w_2$, and the loss still increases strictly on $[0, 1]$. If $z_0 < y/w_2$, continuity keeps $z_\alpha < y/w_2$ on an interval $[0, \alpha_1)$ with $\alpha_1 > 0$. There the derivative is negative and the loss decreases, so it does not increase strictly on $[0, 1]$. This proves the equivalence in (ii).

For $0 < \eta \le (3 - \sqrt{5})/8$, the factors $1 - 2\eta$ and $1 - 4\eta$ are positive, and the bounds above give

$$w_2 z_0 > (1 - 2\eta)(1 - 4\eta) = 1 - 6\eta + 8\eta^2 \ge \tfrac{1}{2},$$

because $8\eta^2 - 6\eta + \frac{1}{2} \ge 0$ for every η at or below its smaller root $(3 - \sqrt{5})/8$. Thus $z_0 > y/w_2$, and the loss increases strictly with α.

The endpoint $\eta = \frac{1}{10}$. At this learning rate, exact rational bounds show that every output z_α passes the target y/w_2. Set

$$c = \frac{1/2}{1/2 + \epsilon} = \frac{1}{1 + 2 \cdot 10^{-8}},$$

so that $0.99999998 < c < 1$. By (A.9),

$$0.9 < w_1 < 0.900000002, \qquad 0.4 < h_1 < 0.40000001, \qquad 0.36 < a_1 < 0.36000001.$$

The Head receives $h_0 = \frac{1}{2}$ and then h_1, and at $\alpha = 1$ each Body coordinate receives $a_0 = \frac{1}{2}$ and then a_1. Their second-step bias-corrected moments therefore satisfy

$$\hat{m}_{2,\mathrm{H}} = \frac{0.9 \cdot \frac{1}{2} + h_1}{1.9} > \frac{17}{38}, \qquad \hat{v}_{2,\mathrm{H}} = \frac{0.999 \cdot (\frac{1}{2})^2 + h_1^2}{1.999} < 0.4528^2,$$
$$\hat{m}_{2,\mathrm{B}} = \frac{0.9 \cdot \frac{1}{2} + a_1}{1.9} > \frac{81}{190}, \qquad \hat{v}_{2,\mathrm{B}} = \frac{0.999 \cdot (\frac{1}{2})^2 + a_1^2}{1.999} < 0.43568^2.$$

The first-moment bounds use $h_1 > 0.4$ and $a_1 > 0.36$. The second-moment bounds follow by substituting the upper bounds on h_1 and a_1 and checking

$$\frac{0.999/4 + 0.40000001^2}{1.999} < 0.4528^2, \qquad \frac{0.999/4 + 0.36000001^2}{1.999} < 0.43568^2,$$

which are inequalities between exact rational numbers. The first step moves the Head and each Body coordinate by ηc. Writing the total moves as ηM_H and ηM_B, with $M = c + \hat{m}_2/(\sqrt{\hat{v}_2} + \epsilon)$,

$$M_\mathrm{H} > 0.99999998 + \frac{17/38}{0.4528 + 10^{-8}} > 1.988,$$
$$M_\mathrm{B} > 0.99999998 + \frac{81/190}{0.43568 + 10^{-8}} > 1.9785,$$

where the final inequalities follow by clearing the positive rational denominators. Hence

$$w_2 = 1 - 0.1 M_\mathrm{H} < 0.8012, \qquad z_1 = 1 - 0.2 M_\mathrm{B} < 0.6043.$$

Since $z_\alpha \le z_1$, and w_2 and z_1 are positive,

$$w_2 z_\alpha \le w_2 z_1 < 0.8012 \cdot 0.6043 = 0.48416516 < \tfrac{1}{2}.$$

The middle factor of (A.17) is therefore negative throughout $[0, 1]$, and the loss decreases strictly with α at this learning rate. □

In this construction a more faithful pair shortens the Body's move along β, and the learning rate decides whether the shorter move helps. For $\eta \le (3 - \sqrt{5})/8$, Reprise's output stops short of the target, so fidelity raises the loss. At $\eta = \frac{1}{10}$ every output passes the target, so fidelity lowers it.